# Aditya Kumar

Fremont, California, United States | adithya.ak@gmail.com | (925) 568-6825 | github.com/adityatheperson | linkedin.com/in/aditya-kumar-uci

## PROFESSIONAL SUMMARY

UC Irvine Computer Science student and first author of an IEEE ICCE paper on video-based ASL translation. Built local QLoRA fine-tuning, model-evaluation, and speech-transcription systems.

## EDUCATION

**University of California, Irvine** Jan 2030
B.S., *Computer Science*

## SKILLS

**Languages:** Python, C++, C#, Kotlin

**Machine learning:** MLX, LoRA/QLoRA, TensorFlow, Keras, scikit-learn, Whisper, Artificial intelligence (AI), AI assistants and agents, AI observability

**Data and evaluation:** NumPy, Pandas, Matplotlib, ROUGE

**Tools:** Git, SQLite, WinUI 3, Jetpack Compose, Codex, Claude

## WORK EXPERIENCE

**Penumbra, Inc. |** Information Technology Intern Jun 2026 - Aug 2026

- Supported endpoint hardware deployment, enterprise software, and troubleshooting for internal users at a medical-device company.

**Leopard Imaging |** Engineering Intern May 2025 - Aug 2025

- Developed and tested Python scripts for camera systems on an AI-enabled imaging team, applying artificial intelligence (AI) and computer vision to real-time image processing.
- Built a stereo-camera depth-estimation prototype using computer vision, turning business requirements into a working solution option and documenting the design for the team.

**Aspiring Scholars Directed Research Program (ASDRP) |** Lead Author, Computer Vision and HCI Research Aug 2023 - Jan 2025

- Built a video-based American Sign Language to English translation system using Python and machine learning, applying AI techniques to a real-world accessibility problem.
- Published as first author in the 2025 IEEE International Conference on Consumer Electronics (ICCE) proceedings, presenting research findings and recommendations to technical audiences.

## PROJECTS

**Local LoRA Summarizer**

- Fine-tuned a LoRA adapter on a 4-bit Qwen3-4B model with QLoRA on Apple MLX for local summarization, prototyping an AI assistant that runs fully on-device; ran 300 training iterations and maintained 24 automated pipeline tests.
- Built a local web evaluation interface comparing base and adapter outputs, using AI observability-style metrics to measure ROUGE-L F1 of 0.4614 (base) and 0.5357 (adapter) on 3 synthetic examples held out for evaluation, a pipeline test rather than evidence of general model quality.

**Zoom Recorder**

- Shipped a Windows app that records Zoom classes, organizes recordings by course, and transcribes audio on device.
- Built local Whisper transcription with a Vulkan GPU worker and explicit CPU fallback; cached models support offline transcription after initial download.
- Automated scheduled class joining with a wake-capable Windows task, overlap warnings, retry-until-cutoff behavior, and readiness checks.